USN | | | | | | | | | | | [23ESEE12]

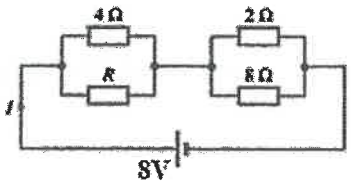
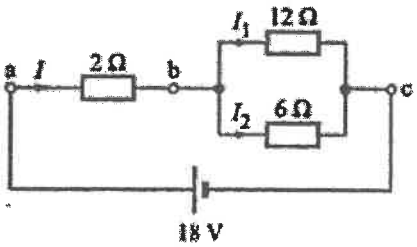
SECOND SEMESTER B.Tech EXAMINATION, JUNE-2024

BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05x02=10**
1.	**a.**	State and explain Kirchhoff's laws.	L2	CO1	**02**
	b.	An alternating current i is given by i = 141.4 sin 314t. Find the frequency of the waveform.	L3	CO2	**02**
	c.	Give the necessity of earthing and its types.	L2	CO3	**02**
	d.	Write down the applications of various types of inductors.	L2	CO5	**02**
	e.	Define the terms : Zener Breakdown , Avalanche Breakdown	L2	CO6	**02**
		PART - B			**06x05=30**
2.	**a.**	Find the value of R for the circuit shown in Fig.1 if the total power dissipated for the circuit is 16W.  Fig.1	L3	CO1	**05**
		OR			
	b.	Determine the unknowns using current divider rule shown in Fig.2  Fig.2	L3	CO1	**05**
3.	**a.**	Prove that the power consumed by pure inductor is zero.	L3	CO2	**05**
		OR			
	b.	Derive RMS value of sinusoidal voltage in term of its maximum value.	L3	CO2	**05**
4.	**a.**	Discuss in brief on the different types of switches used in Electrical circuits.	L2	CO3	**05**
		OR			
	b.	Discuss in brief on the different types of batteries used in Electrical circuits.	L2	CO3	**05**

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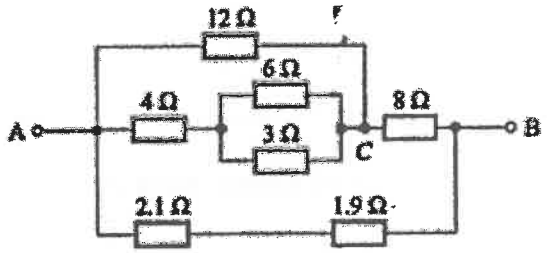
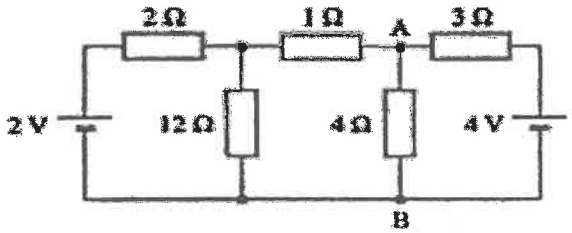
5.	**a.**	When two resistance R1 & R2 are connected in parallel, they dissipate four times the power that they dissipate when they are connected in series with the same ideal source of emf. If R1=3Ω, find R2.	L3	CO4	**05**
		OR			
	b.	Two bulbs of 40 W and 60 W are connected in series to an external potential difference. Which bulb will glow brighter? Why?	L3	CO4	**05**
6.	**a.**	List the various capacitors used in electronic circuits and state uses of each type.	L2	CO5	**05**
		OR			
	b.	Write short notes on wire wound resistors.	L2	CO5	**05**
7.	**a.**	Describe the action of PN junction diode under forward bias and reverse bias conditions.	L2	CO6	**05**
		OR			
	b.	Show that the zener diode can be used as a voltage regulator.	L2	CO6	**05**
		PART - C		**06x10=60**	
8.	**a.**	Determine total resistance across AB and also find the total current if a 100V is applied across AB shown in Fig.3.  Fig.3	L3	CO1	**10**
		OR			
	b.	Find all mesh currents and determine the current through 4Ω through A to B for the given circuit Fig.4 shown below.  Fig.4	L3	CO1	**10**
9.	**a.**	A capacitive reactance of 40 Ω is connected in series with a non-inductive resistance of 30Ω across a 100V, 50Hz supply. Find **i.** Impedance **ii.** Current	L3	CO2	**10**

		iii. Phase angle **iv.** Equation for the instantaneous value of current **v.** Capacitance			
		OR			
	b.	Derive the expression for voltage, current and power for a single-phase AC circuit containing RLC in series. Draw the necessary waveform and phasor diagrams.	L3	CO2	**10**
10.	**a.**	Write a short note on MCB and Switch Fuse Unit (SFU).	L2	CO3	**10**
		OR			
	b.	Explain how a lamp is controlled using two way and three- way connections with neat circuits.	L2	CO3	**10**
11.	**a.**	The cost of boiling 2 kg of water in an electric kettle is 12 paise. The kettle takes 6 minutes to boil water from an ambient temperature of 20°C, calculate **i.** the efficiency of the kettle and **ii.** wattage or power rating of the kettle if cost of 1 kWh is 40 paise.	L3	CO4	**10**
		OR			
	b.	In a circuit, the applied voltage of 150 V lags, the current of 8 A by 40° **i.** Is p.f. lagging or leading? **ii.** What is the value of p.f.? **iii.** Is the circuit inductive or capacitive? **iv.** What is the value of active and reactive power? **v.** Calculate apparent power.	L3	CO4	**10**
12.	**a.**	Draw and Explain the working of regulated power supply with neat block diagram.	L2	CO5	**10**
		OR			
	b.	With the suitable block diagram explain the basic instrumentation system.	L2	CO5	**10**
13.	**a.**	Explain the action of a full-wave bridge rectifier and give wave forms of input and output voltages.	L2	CO6	**10**
		OR			
	b.	Draw the circuit diagram of an half wave rectifier with and without filter, and ex plain its operation and waveforms.	L2	CO6	**10**